

ITCS 3166

Assignment 1

1. (15 pts) The performance of a client-server system is strongly influenced by two major network characteristics: the bandwidth of the network (that is, how many bits/sec it can transport) and the latency (that is, how many seconds it takes for the first bit to get from the client to the server). Give an example of a network that exhibits high bandwidth but also high latency. Then give an example of one that has both low bandwidth and low latency.

A transcontinental fiber link might have many gigabits/sec of bandwidth, but the latency will also be high due to the speed of light propagation over thousands of kilometers. In contrast, a 56-kbps modem calling a computer in the same building has low bandwidth and low latency.

2. (15 pts) What are two reasons for using layered protocols? What is one possible disadvantage of using layered protocols?

Among other reasons for using layered protocols, using them leads to breaking up the design problem into smaller, more manageable pieces, and layering means that protocols can be changed without affecting higher or lower ones. One possible disadvantage is the performance of a layered system is likely to be worse than the performance of a monolithic system, although it is extremely difficult to implement and manage a monolithic system.

3. (10 pts) What is the main difference between TCP and UDP?

TCP is connection oriented, whereas UDP is a connectionless service.

4. (20 pts) An image is 1600×1200 pixels with 3 bytes/pixel. Assume the image is uncompressed. How long does it take to transmit it over a 56-kbps modem channel? Over a 1-Mbps cable modem?

The image is $1600 \cdot 1200 \cdot 3$ bytes or 5,760,000 bytes. This is 46,080,000

bits. At 56,000 bits/sec, it takes about 822.857 sec. At 1,000,000 bits/sec, it takes 46.080 sec. At 10,000,000 bits/sec, it takes 4.608 sec.

5. (20 pts) Calculate the end-to-end transit time for a packet for both GEO (altitude: 35,800 km), MEO (altitude: 18,000 km) and LEO (altitude: 750 km) satellites.

Transit time by GEO= $(35800*2)/300000=0.24\text{sec}$

Transit time by MEO= $(18000*2)/300000=0.12\text{sec}$

Transit time by LEO= $(750*2)/300000=0.005\text{sec}$

6. (20 pts) Three packet-switching networks each contain n nodes. The first network has a star topology with a central switch, the second is a (bidirectional) ring, and the third is fully interconnected, with a wire from every node to every other node. What are the best-, average-, and worst-case transmission paths in hops?

star: 2, 2, 2 hops (best, average, worst)

ring: 1, $n/4$, $n/2$

full: 1, 1, 1